

Griffin Flight Hour Predictor



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PROBLEM STATEMENT: How can the U.S. Army better estimate future flight requirements by applying analytic models to historical, unit-specific flight data at the battalion and above level, compared to manual or existing processes?

Current Process and Financial Impacts

Eleven-step, manual process.

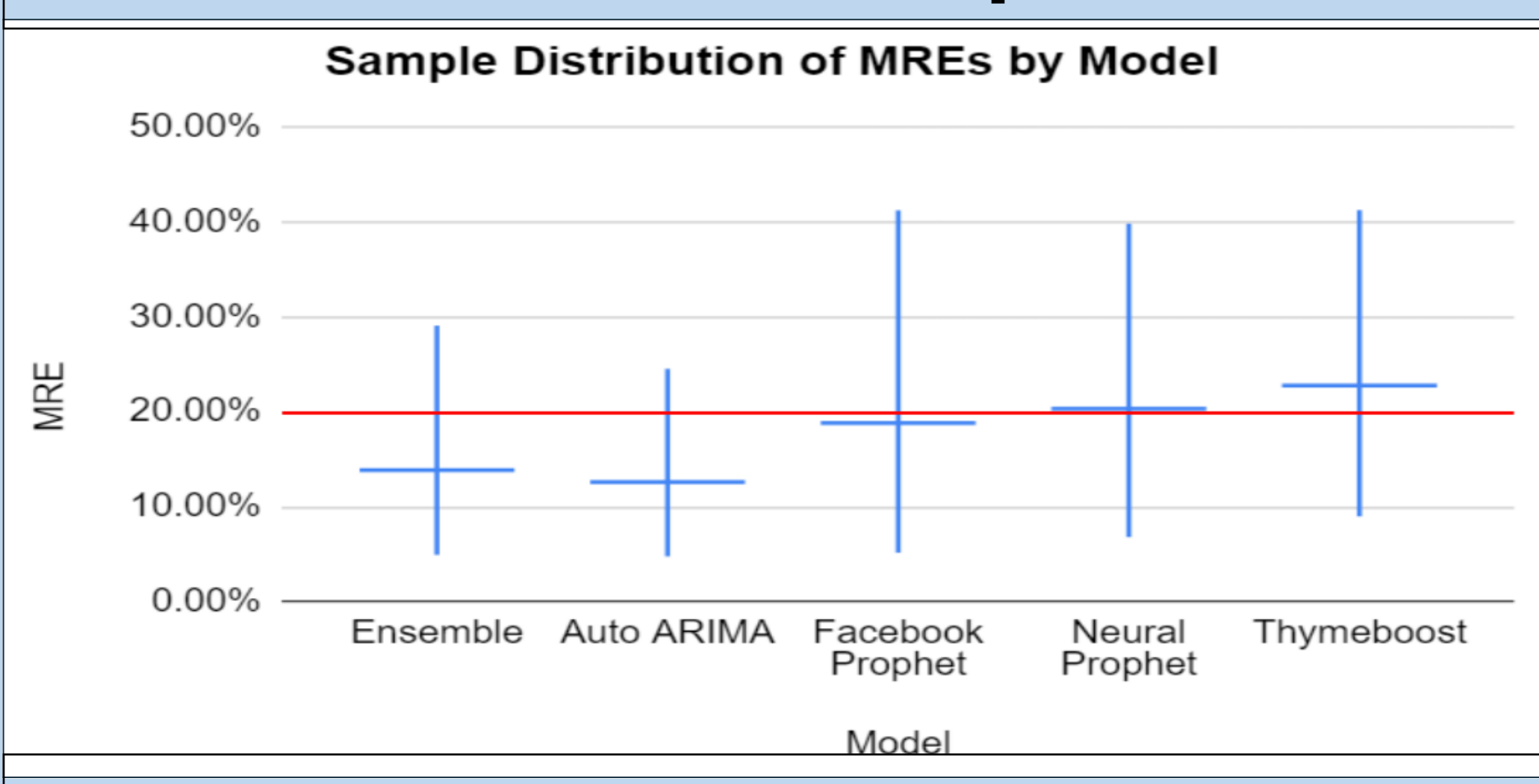
FORMULATING A UNIT FLYING-HOUR PROGRAM			
9-3. Unit	Qualification Training RL 3 to RL 1 RL 2 to RL 1 FAC 1 Aviator Continuation FAC 2 Aviator Continuation	Step 4. Determine the total number of hours required for assigned crewmembers to attain and maintain RL 1 for the training year by adding the progression hour total (step 5) to the continuation hour total (step 6).	Aviator progression hours Aviator continuation hours Unit METL task proficiency hours
Assignee	Step 5. Calculate the Factor the Below, two training. To be designated	Step 9. Support hours. Determine mission support requirements such as CTC rotations, FTXs, and so forth. This can be a block hour commitment or the number of aircraft required by hours per aircraft. Determine the number of total training hours (step 7) that can be accomplished concurrently during mission support. (This number is derived by historical analysis.) Subtract this number from the total support hours required. The result will be the additional hours that should be added by this step to the total unit flying-hour requirement.	724 hours 2055 hours 300 hours Total training hours: 3079 hours
Annual	Qualification Training RL 3 to RL 1 RL 2 to RL 1	Step 10. Maintenance hours. Multiply the unit cumulative hour amount (sum of step 7 and step 8) by 5 percent to determine the estimated maintenance-hour requirements. The 5 percent figure is a general guideline and units should determine their own multiplier based on previous experience and forecasted maintenance.	National Training Center rotation Infantry support SOF support Division FTX Total support hours: 2400 hours 1900 hours 500 hours Subtract training hours conducted concurrently: Additional hours:
70 X 33	Step 6. Calculate the requirement and those of FAC 1 and Take into progress to	Step 11. Total flying-hour requirement. Add the unit cumulative hour amount to the estimated maintenance hours (each determined in step 9) to determine the overall unit flying-hour requirement.	Total training hours Additional hours Unit cumulative hours Maintenance hours Unit Flying Hour Requirement:
	FAC 1 Aviators, RL 1 en FAC 2 Aviators, RL 1 en Newly assigned FAC 1 A Newly assigned FAC 2 A * 66% = 8 months of the the average time new Aviators are estimated		3079 hours +500 hours 3579 hours x .05 179 maintenance hours 3579 hours +179 hours 3758 hours

Unit's Cost Variance, by Accuracy

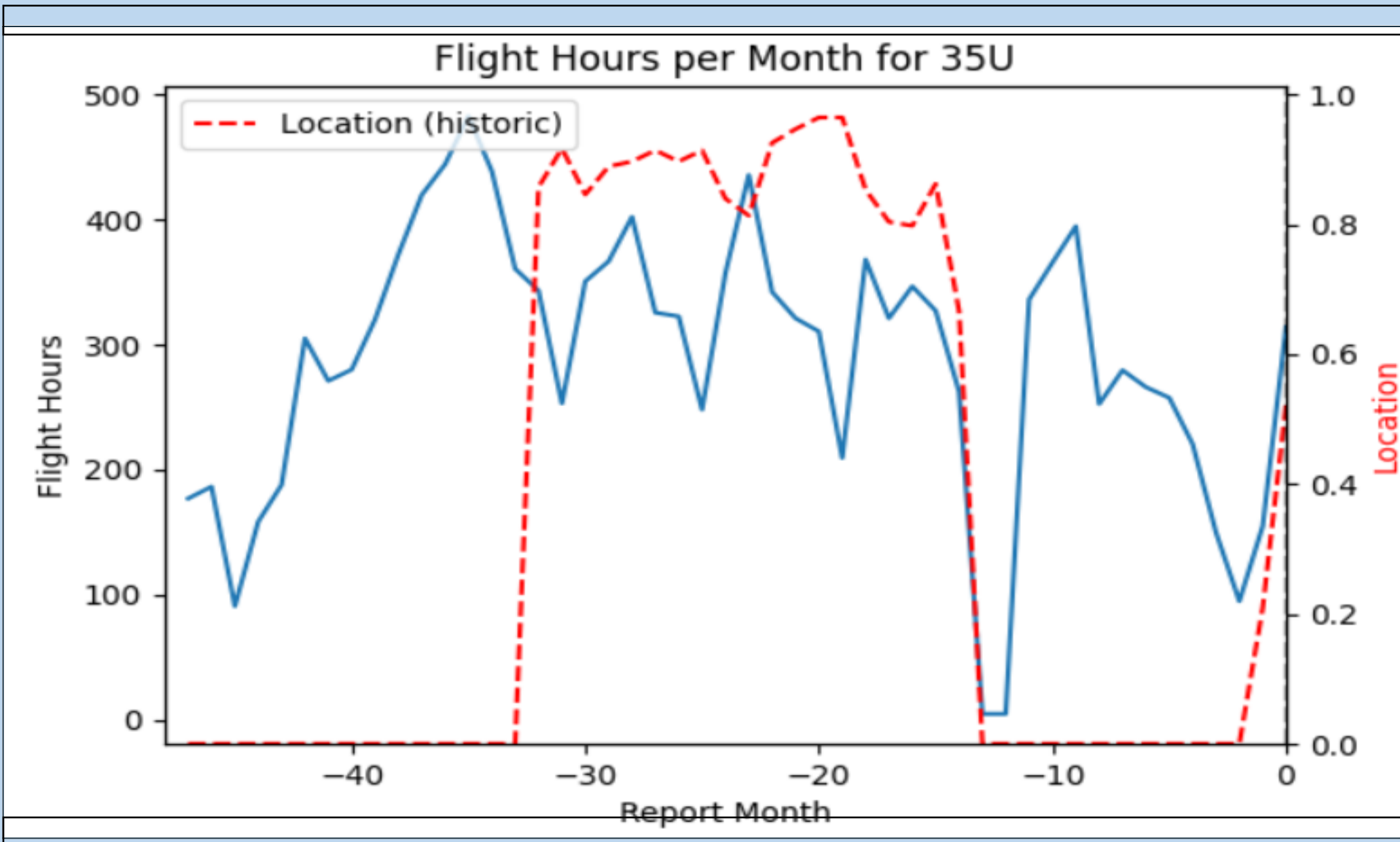
Method	Variance	Cost Variance (mil)
Benchmark	0.20	+/- \$1.750
Griffin (avg)	0.13	+/- \$1.140
Griffin (best)	0.02	+/- \$0.175

Sample Attack aviation battalion flying 3,384 hours, resulted in \$17.5 million operating budget.

Solution Development

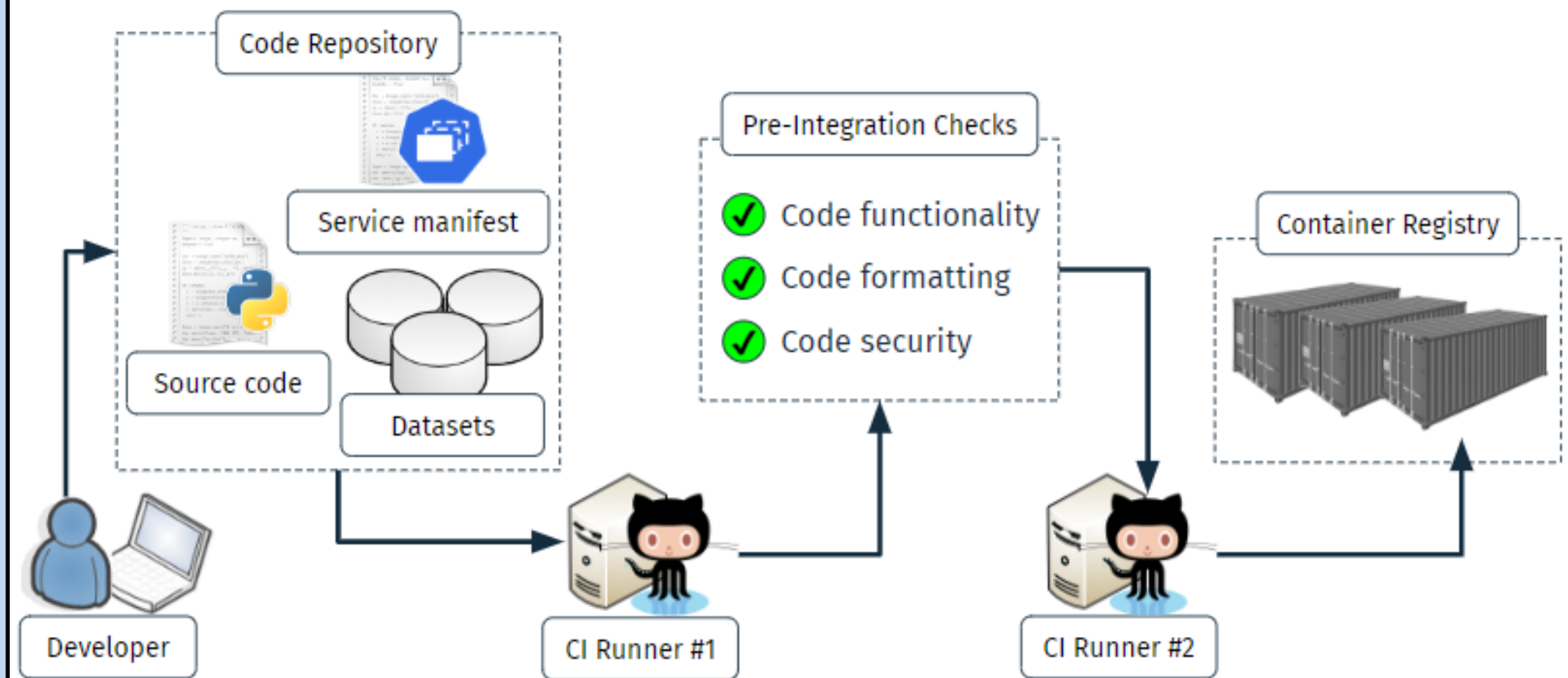


Candidate Model Architecture Comparisons



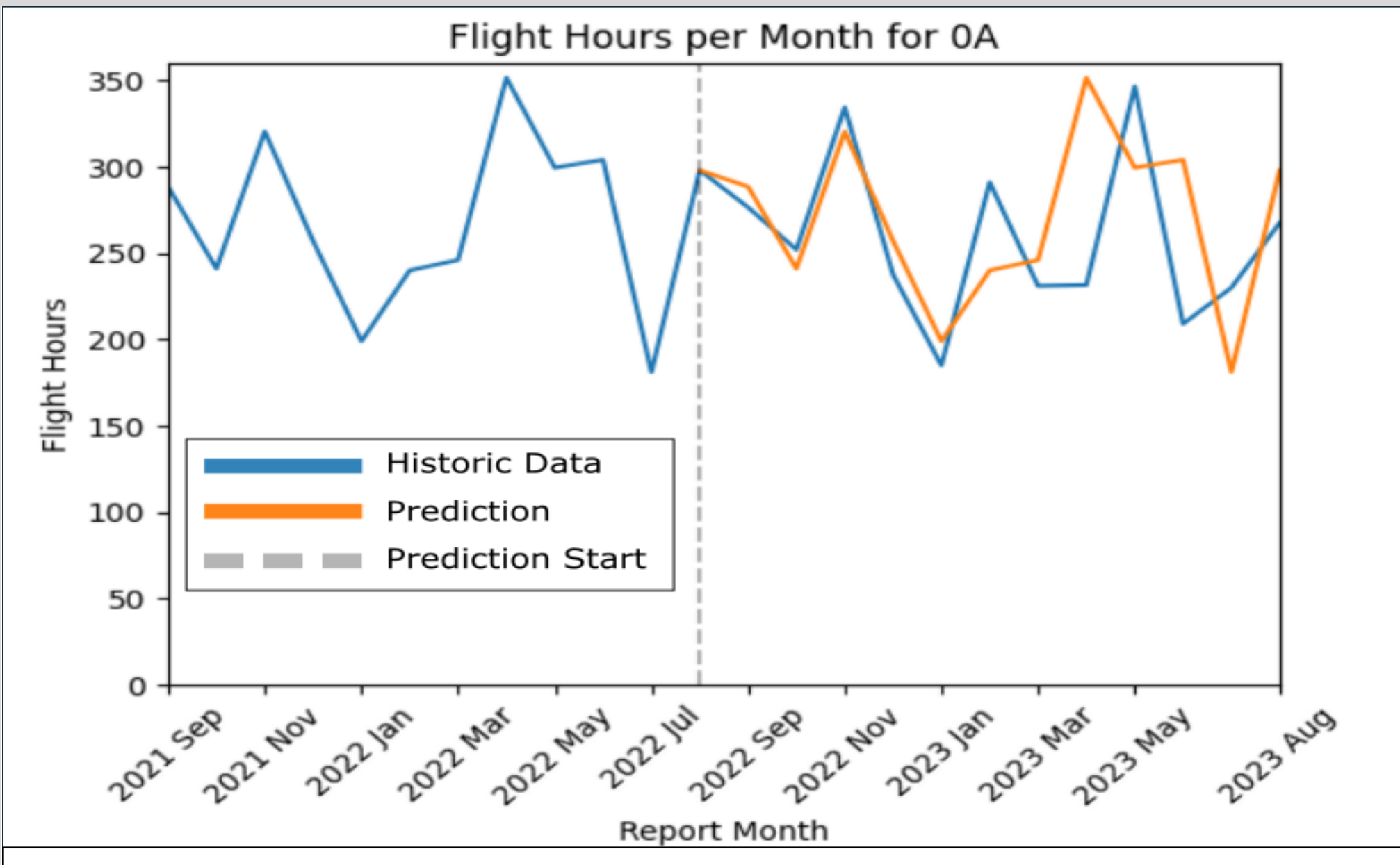
Attempted Feature Engineering for Location

DevSecOps: Continuous Integration (CI) Process

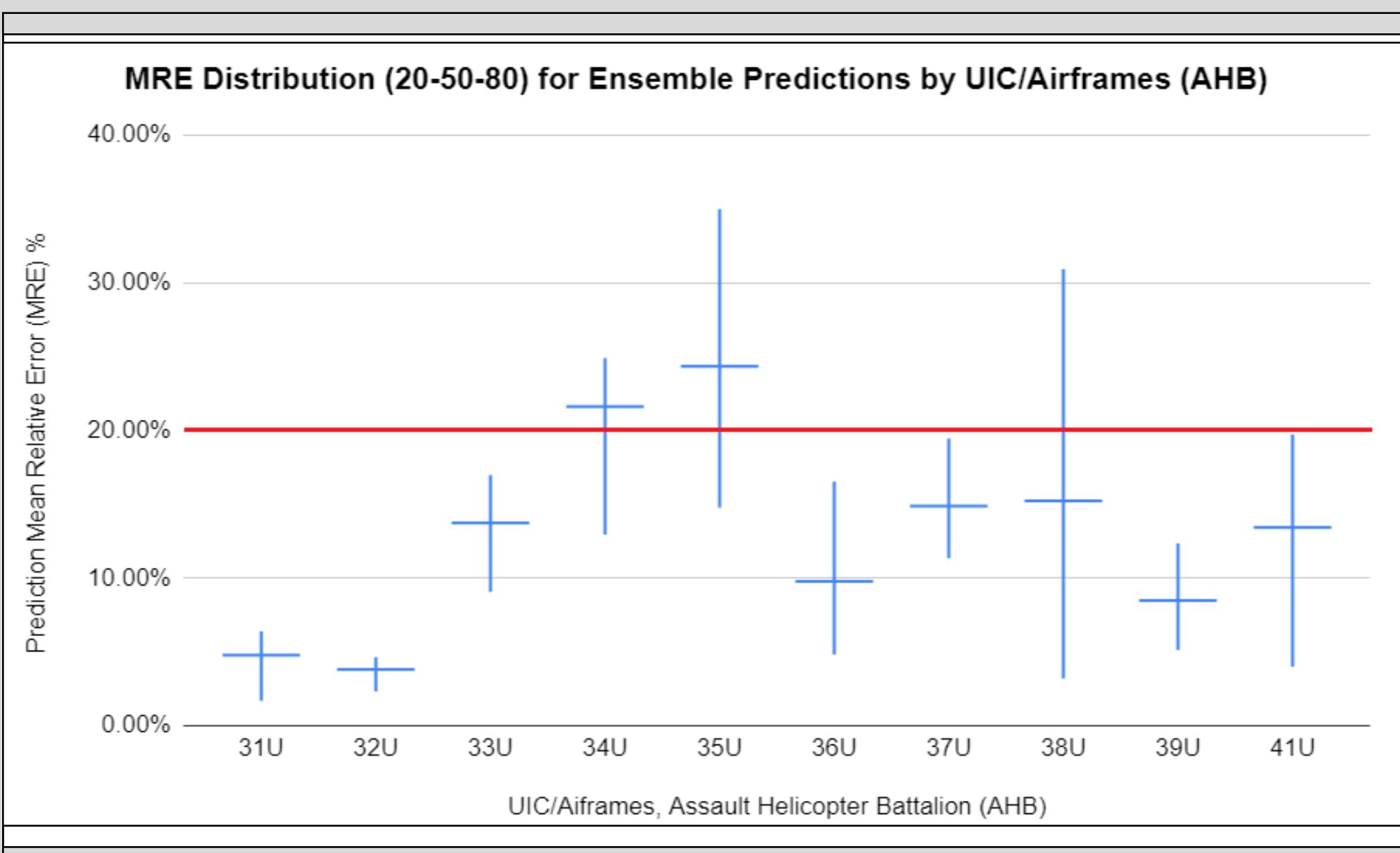


CI Pipeline Processes Enabled Team Collaboration

Individual Unit Results



Sample Monthly Prediction



Results by Unit and Aircraft Type

CONCLUSION: An ensemble model, combining four time-series machine learning models, was the most accurate and consistent, compared to individual models.

Team achieved **overall 13% mean relative error** for predictions, with 25 of 41 organizations consistently below the 20% mean relative error goal.